THE PUSHFORWARD ALONG A FINITE ETALE MORPHISM PRESERVES CONSTRUCTIBILITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three settings of the statement.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf and finite locally constant sheaf. Same chapter, the definitions of constructibility and of finite local constancy, and the characterisation of constructibility by a partition into constructible locally closed strata. Supplies the objects and the criterion used in the proof.
- Pushforward along a finite etale morphism and its exactness. Same chapter, the description of the pushforward along a finite etale morphism by the product over the fibres, which is exact and computes the pushforward of a locally constant sheaf as again locally constant. Supplies the computation showing that the partition is preserved.

Source reference. Original lemma 095F: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11789–11789. Navigation pointer invoked at proof line 12330; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095Q: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12276–12276. Navigation pointer invoked at proof line 12331; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 04HN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12335–12335. Navigation pointer invoked at proof line 12335; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a finite étale morphism of schemes. Let Λ be a Noetherian ring. If $\mathcal{F}$ is a constructible sheaf of sets, constructible sheaf of abelian groups, or constructible sheaf of Λ-modules on $X_{étale}$, the same is true for $f_*\mathcal{F}$ on $Y_{étale}$.*

Proof. By Lemma 095F it suffices to check this Zariski locally on Y and by Lemma 095Q we may replace Y by an étale cover (the construction of f_* commutes with étale localization). A finite étale morphism is étale locally isomorphic to a disjoint union of isomorphisms, see Étale Morphisms, Lemma 04HN. Thus, in the case of sheaves of sets, the lemma says that if $\mathcal{F}_i$, $i = 1, \ldots, n$ are constructible sheaves of sets, then $\prod_{i=1,\ldots,n} \mathcal{F}_i$ is too. This is clear. Similarly for sheaves of abelian groups and modules. □